

ZAKOPANE, Poland

WMO: 126250

Lat: 49.2939N

Lon: 19.9603E

Elev: 857

StdP: 91.44

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-14.0	-19.1	0.8	-15.7	-16.5	1.0	-13.0	6.9	3.2	6.0	3.2	0.8	170	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	26.7	17.8	24.9	17.1	23.2	16.3	18.8	24.5	17.9	23.3	17.0	22.1	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.2	21.1	15.9	12.5	20.3	15.0	11.9	19.0	57.0	24.9	53.9	23.3	51.2	22.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.6	3.9	-19.7	29.6	3.5	1.6	-22.2	30.7	-24.2	31.6	-26.2	32.5	-28.7	33.6		
(5)			-20.1	20.3	3.2	1.0	-22.4	21.0	-24.2	21.6	-26.0	22.2	-28.3	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-3.7	-2.6	0.7	6.4	10.7	14.2	15.9	15.6	11.2	7.0	2.8	-2.0
	DBStd	8.16	5.06	5.22	4.73	4.27	3.76	3.55	3.16	3.22	3.62	4.45	4.98	5.06
	HDD10.0	1966	426	352	288	122	37	6	1	2	29	114	219	372
	HDD18.3	4390	684	585	546	358	238	131	88	97	216	352	466	630
	CDD10.0	651	0	0	0	13	59	131	183	176	64	21	4	0
	CDD18.3	32	0	0	0	0	1	7	12	12	1	0	0	0
	CDH23.3	348	0	0	0	1	6	74	127	131	9	1	0	0
	CDH26.7	50	0	0	0	0	0	8	18	22	1	0	0	0
Wind	WSAvg	1.5	1.5	1.7	1.8	1.7	1.6	1.3	1.3	1.2	1.2	1.4	1.6	1.6
Precipitation	PrecAvg	1120	44	44	54	83	124	169	176	144	97	68	65	54
	PrecMax	1436	127	95	139	167	240	325	415	323	247	228	185	111
	PrecMin	811	8	4	1	24	48	67	55	38	17	13	25	6
	PrecStd	152	25	24	26	32	45	62	84	66	48	45	32	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	11.2	14.7	21.5	24.0	27.8	28.7	29.2	24.4	21.0	16.3	11.3
		MCWB	4.9	4.8	7.1	11.9	14.7	18.5	18.4	18.2	15.9	13.3	9.7	7.1
	2%	DB	7.6	8.6	12.2	18.9	22.1	25.5	26.6	26.5	22.1	18.5	14.1	9.1
		MCWB	4.6	3.9	6.4	10.9	14.1	17.4	17.9	18.0	15.3	12.3	9.2	5.4
	5%	DB	5.4	6.7	10.2	16.3	20.3	23.4	24.5	24.5	20.1	16.6	12.5	7.1
		MCWB	3.2	3.4	5.5	9.3	13.4	16.4	17.3	17.2	14.4	11.5	8.4	4.1
	10%	DB	3.1	4.6	8.0	14.0	18.2	21.4	22.7	22.5	18.2	14.6	10.6	5.2
		MCWB	1.5	2.1	4.2	8.4	12.7	15.4	16.4	16.4	13.3	10.5	7.3	3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.9	8.2	12.5	16.4	19.6	20.0	19.6	17.0	14.3	11.0	7.2
		MCDB	8.1	9.4	12.6	19.9	21.7	25.5	25.7	25.6	22.4	19.0	14.5	10.4
	2%	WB	4.9	4.7	7.0	11.2	15.1	18.1	18.8	18.6	15.9	13.0	9.7	6.0
		MCDB	7.4	7.9	11.1	17.7	19.9	23.8	24.2	24.6	20.5	17.2	13.4	8.5
	5%	WB	3.2	3.6	5.8	10.1	14.0	16.9	17.9	17.7	14.8	11.9	8.6	4.5
		MCDB	5.2	6.3	9.5	15.3	18.7	22.0	23.0	23.4	19.0	15.7	12.0	6.6
	10%	WB	1.7	2.3	4.6	8.8	13.0	15.9	17.0	16.8	13.8	10.7	7.4	3.0
		MCDB	3.1	4.4	7.7	13.1	17.8	20.4	21.9	21.8	17.5	14.2	10.4	4.9
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	8.1	9.5	9.5	9.4	9.6	10.1	9.1	9.0	7.4	6.9
		MCDBR	8.5	10.1	11.6	13.3	13.5	13.3	13.2	13.3	12.6	12.6	10.1	8.5
		MCWBR	6.0	6.6	6.9	6.7	6.9	6.5	6.2	6.2	6.9	7.4	6.2	5.8
	5% WB	MCDBR	8.0	9.0	10.5	11.9	12.1	11.8	11.5	12.2	11.2	12.0	9.0	7.9
MCWBR		5.8	6.2	6.6	6.5	6.6	6.2	6.1	6.3	6.5	7.4	5.9	5.9	
Clear-Sky Solar Irradiance	taub	0.286	0.313	0.360	0.420	0.416	0.418	0.431	0.425	0.392	0.355	0.323	0.284	
	taud	2.285	2.219	2.172	2.124	2.200	2.234	2.199	2.220	2.279	2.357	2.371	2.342	
	Ebn at Noon	762	823	838	822	844	844	825	810	799	767	711	714	
	Edh at Noon	71	97	121	142	137	134	137	128	109	85	67	59	
All-Sky Solar Radiation	RadAvg	0.91	1.66	2.79	4.07	4.89	5.22	5.06	4.64	3.27	2.00	1.02	0.70	
	RadStd	0.14	0.25	0.42	0.47	0.62	0.49	0.58	0.41	0.48	0.28	0.14	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		+0.57	N/A	N/A	+0.95	+0.50	+0.43	-130	-184	+88	+38

Nomenclature: See separate page